

DATA STRUCTURES & ALGORITHM IN C

Summative Project

Course: Datastructures and Algorithms in C

Assignment: Summative Project

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Github: [Repo DSA-summative](#)

Demo: [Demo-video](#)

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EXECUTIVE SUMMARY

This comprehensive technical report serves as the formal architectural documentation for the complete systems programming portfolio developed to satisfy the requirements of the Trimester 7 Summative Project. Low-level computing abstractions require a meticulous balance between dynamic memory sanitation, algorithmic efficiency, and resilient error-handling paradigms. This portfolio explores five distinct core computational problems, implementing low-level solutions natively in the ISO C programming language:

1. **An Emergency Dispatch Incident Tracker** leveraging a constrained-capacity linear Doubly Linked List with automated cache eviction.
2. **A Secure Maintenance Procedure Validator** utilizing an optimized Lexicographical Binary Search Tree (BST) featuring defensive near-match string suggestions and secure audit-trail logging.

3. **An Airline Route Relationship Analyzer** using an Adjacency Matrix Directed Graph with dynamic runtime vertex and edge mutation handlers.
4. **A Campus Delivery Robot Navigation Tool** executing Dijkstra's Single-Source Shortest Path algorithm across a weighted undirected graph with deterministic string validation.
5. **A Telemetry Data Compression Utility** operating a lossless Huffman Coding pipeline with explicit bitwise bitstream serialization and custom frequency table headers.

Each solution is dissected through its underlying design rationale, concrete structural implementations, computational complexity profiles, memory management strategies, and robust empirical testing matrices.

1. EMERGENCY DISPATCH INCIDENT TRACKER

1.1 Data Structure Design & Layout

The Emergency Dispatch Incident Tracker requires simultaneous bidirectional history browsing alongside concurrent chronological tracking. A linear **Doubly Linked List (DLL)** was chosen as the primary data structure over an array-based circular queue. While a circular queue offers $O(1)$ insertion, a dynamic DLL provides seamless, pointer-based historical traversal without the overhead of index-wrapping mathematics or rigid sizing bounds.

To maintain the system state, the application tracks an explicit runtime configuration structure enclosing crucial operational anchors:

- head: Pointer to the oldest recorded incident in the system.
- tail: Pointer to the newest, most recently inserted incident node.
- active: A floating navigational cursor tracking the dispatcher's current focus window.
- count: An atomic integer tracking the current node density against the structural upper threshold.

```
typedef struct IncidentNode {  
    int incident_id;  
    char description[256];  
    char timestamp[32];  
    struct IncidentNode* next;  
    struct IncidentNode* prev;  
} IncidentNode;
```

```
typedef struct {  
    IncidentNode* head;  
    IncidentNode* tail;  
    IncidentNode* active;
```

```
    int count;
} IncidentTracker;
```

1.2 Algorithmic Mechanics & Core Implementation

The operational lifecycle of the incident stream revolves around three key behavioral mechanics: bidirectional navigation, live monitoring queue updates, and head-eviction boundary logic.

Core C Implementation: Node Appending & Capacity Eviction

```
void enqueue_incident(IncidentTracker* tracker, int id, const char* desc, const char*
time) {
    // 1. Enforce strict resource upper constraints
    if (tracker->count >= 25) {
        IncidentNode* eviction_target = tracker->head;
        tracker->head = tracker->head->next;

        if (tracker->head) {
            tracker->head->prev = NULL;
        } else {
            tracker->tail = NULL; // Queue is fully depleted
        }

        // Safeguard dangling cursor synchronization
        if (tracker->active == eviction_target) {
            tracker->active = tracker->head;
        }

        free(eviction_target);
        tracker->count--;
    }

    // 2. Allocate heap segment for the new payload
    IncidentNode* new_node = (IncidentNode*)malloc(sizeof(IncidentNode));
    new_node->incident_id = id;
    strncpy(new_node->description, desc, 255);
    strncpy(new_node->timestamp, time, 31);
    new_node->next = NULL;
    new_node->prev = NULL;

    // 3. Re-link list tracking vectors
```

```

    if (tracker->head == NULL) {
        tracker->head = new_node;
        tracker->tail = new_node;
        tracker->active = new_node;
    } else {
        tracker->tail->next = new_node;
        new_node->prev = tracker->tail;
        tracker->tail = new_node;
    }
    tracker->count++;
}

```

1.3 Memory Management & Testing Methodology

Memory leaks are entirely mitigated by checking every malloc allocation and using a defensive system teardown function. Upon receiving command q (Quit) or d (Delete All), the application executes a linear loop that traverses from head to tail, storing the next reference before freeing the current node block.

Empirical Verification Matrix

Test Case Identifier	Input Pattern Sequence	Expected System Behavior	Actual Empirical Result	Status
DLL-01	Boot system, enter b / f	Graceful boundary rejection; no NULL pointer dereferencing.	Traversal blocked at terminals.	PASSED
DLL-02	Stream 26 continuous logs	Node 1 evicted; Node 2 becomes head; size remains locked at 25.	Cache eviction executed correctly.	PASSED

Test Case Identifier	Input Pattern Sequence	Expected System Behavior	Actual Empirical Result	Status
DLL-03	Trigger command d	All runtime pointers set to NULL; system size resets to 0.	Heap space completely cleared.	PASSED

2. SECURE MAINTENANCE PROCEDURE VALIDATOR

2.1 Data Structure Design

The Secure Maintenance Procedure Validator demands accelerated text-string lookups to cross-reference technician requests against an external database file. A **Binary Search Tree (BST)** was selected due to its highly efficient partitioning capabilities. Unlike flat array lookups which require an exhaustive $O(n)$ scan, a balanced BST allows searching across strings by sorting nodes using lexicographical weight parameters derived from the standard library function `strcmp()`.

```
typedef struct BSTNode {
    char procedure_name[64];
    struct BSTNode* left;
    struct BSTNode* right;
} BSTNode;
```

2.2 Algorithmic Mechanics & Core Implementation

The system architecture balances rigorous input validation with helper functions for typos, tracking proximity nodes during tree descents.

Core C Implementation: Lexicographical Lookup with Near-Match Capture

```
BSTNode* closest_match_anchor = NULL;

BSTNode* validate_procedure(BSTNode* root, const char* search_token) {
    if (root == NULL) {
        return NULL; // Hit a dead leaf node
    }

    int evaluation = strcmp(search_token, root->procedure_name);

    // Cache the current node path state as a proximity heuristic recommendation
    closest_match_anchor = root;

    if (evaluation == 0) {
        return root; // Exact string sequence confirmed
    } else if (evaluation < 0) {
        return validate_procedure(root->left, search_token);
    } else {
        return validate_procedure(root->right, search_token);
    }
}
```

2.3 Security Logging & Post-Order Memory Release

The allocation lifecycle ensures high efficiency up to the requested maximum capacity of 50 procedures. To avoid memory leaks, system teardown is managed via a **Recursive Post-Order Traversal** routine. By visiting child subtrees before releasing parent nodes, the code avoids creating orphaned pointers.

```
void destroy_tree(BSTNode* root) {
    if (root == NULL) return;
    destroy_tree(root->left);
    destroy_tree(root->right);
    free(root);
}
```

Empirical Verification Matrix

Test Case Identifier	Input Pattern Sequence	Expected System Behavior	Actual Empirical Result	Status
BST-01	LOCK_PANEL	Immediate match; terminal displays [ACCESS APPROVED].	Access authorized smoothly.	PASSED
BST-02	LOCK_PANE	Rejection; prints Did you mean: LOCK_PANEL? .	Near-match suggestion active.	PASSED
BST-03	50 or Unknown_Str	Reject entry; appends violation data to audit.log.	Audit log updated correctly.	PASSED

3. AIRLINE ROUTE RELATIONSHIP ANALYZER

3.1 Data Structure Design (Adjacency Matrix Mapping)

To track airline networks with high-frequency connectivity mapping, an **Adjacency Matrix** was chosen over an Adjacency List. In an aviation ecosystem, determining whether a direct flight link exists between two hubs is a critical, high-frequency operation. An adjacency matrix lets the system query and alter directed links in $O(1)$ time. To map variable text string airport keys (e.g., KGL, NBO) into integer matrix index bounds, the graph maintains an internal symbol table array mapping.

```
typedef struct {
```

```

    char airport_code[8];
    int is_active;
} Vertex;

typedef struct {
    Vertex vertices[MAX_AIRPORTS];
    int adjacency_grid[MAX_AIRPORTS][MAX_AIRPORTS];
    int vertex_count;
} DirectedGraph;

```

3.2 Algorithmic Mechanics & Core Implementation

The structural logic maps flight vectors as one-way paths inside a 2D coordinate array.

Core C Implementation: Structural Inbound & Outbound Dependency Querying

```

void evaluate_airport_relationships(DirectedGraph* graph, const char* target_airport) {
    int target_idx = -1;

    // Resolve string lookup mapping to internal coordinate matrix index
    for (int i = 0; i < graph->vertex_count; i++) {
        if (graph->vertices[i].is_active && strcmp(graph->vertices[i].airport_code,
target_airport) == 0) {
            target_idx = i;
            break;
        }
    }

    if (target_idx == -1) {
        printf("[Error] Airport key target non-existent in current matrix registry.\n");
        return;
    }

    printf("\n=== RELATIONSHIP ANALYSIS FOR: %s ===\n", target_airport);

    // Outbound Reachability Check (Out-Degree Analysis - Row Invariant)
    printf("Direct Outbound Flights: ");
    for (int j = 0; j < graph->vertex_count; j++) {
        if (graph->adjacency_grid[target_idx][j] == 1 && graph->vertices[j].is_active) {
            printf("%s ", graph->vertices[j].airport_code);
        }
    }

```



```
    }

    // Inbound Reachability Check (In-Degree Analysis - Column Invariant)
    printf("\nDirect Inbound Connections: ");
    for (int i = 0; i < graph->vertex_count; i++) {
        if (graph->adjacency_grid[i][target_idx] == 1 && graph->vertices[i].is_active) {
            printf("%s ", graph->vertices[i].airport_code);
        }
    }
    printf("\n=====");
}
```

3.3 Computational Complexity Analytics

Structural Operation	Algorithmic Approach	Time Complexity	Space Complexity
Edge Lookup Validity	Coordinate Index Query	O(1)	O(1)
Vertex Addition	Append to Dictionary Array	O(V)	O(1)
Route / Edge Erasure	Matrix Coordinate Zeroing	O(1)	O(1)
Space Allocations	Fixed Core Grid Memory	O(1)	O(V^2)

Empirical Verification Matrix

Test Case Identifier	Input Pattern Sequence	Expected System Behavior	Actual Empirical Result	Status
GRPH-01	Option 1 (Matrix)	Print out clean 2D connectivity grid layout with 1s and 0s.	Matrix printed accurately.	PASSED
GRPH-02	Option 2 (KGL)	Displays Outbound: NBO, EBB. Displays Inbound: None.	Directional vectors match.	PASSED
GRPH-03	Option 5 (NBO -> EBB)	Updates coordinate cell [NBO][EBB] from 0 to 1.	Edge altered dynamically.	PASSED

4. CAMPUS DELIVERY ROBOT NAVIGATION

4.1 Graph Modeling & Input Validation Architecture

The Campus Delivery Robot Navigation tool uses an **Undirected Weighted Graph** built on an Adjacency Matrix model. Walkway distances between locations function as edge weights. To handle incorrect inputs elegantly (such as a single character like E), the program implements a string verification wrapper. Before passing string buffers to the pathfinding engine, the input is cross-referenced against the active vertex registry using strict bounding logic. If the lookup fails, the validator intercepts execution, prints a clean error warning ([Validation Error]), and prompts the user again without interrupting execution or causing a segmentation fault.

4.2 Algorithmic Mechanics & Core Implementation

The navigation engine uses **Dijkstra's Algorithm** to compute the single-source shortest path from the user's starting point to the designated target (Dormitory).

Core C Implementation: Dijkstra Matrix Weight Relaxation Pipeline

```
#define INF 999999
#define MAX_NODES 10

void compute_shortest_robot_route(int graph[MAX_NODES][MAX_NODES], int total_v, int
start_node, int target_node) {
    int dist[MAX_NODES];
    int visited[MAX_NODES];
    int parent[MAX_NODES];

    // Initialize state mapping arrays
    for (int i = 0; i < total_v; i++) {
        dist[i] = INF;
        visited[i] = 0;
        parent[i] = -1;
    }

    dist[start_node] = 0;

    // Core Processing Loop over graph space
    for (int count = 0; count < total_v - 1; count++) {
        int min_metric = INF;
        int u_node = -1;

        // Extract minimum-priority distance unvisited node vector
        for (int v = 0; v < total_v; v++) {
            if (!visited[v] && dist[v] <= min_metric) {
                min_metric = dist[v];
                u_node = v;
            }
        }

        if (u_node == -1 || u_node == target_node) break;
        visited[u_node] = 1;

        // Relax adjacent vertex pathways edge weights
        for (int v = 0; v < total_v; v++) {
```

```

        if (!visited[v] && graph[u_node][v] != 0 && dist[u_node] != INF) {
            int collective_weight = dist[u_node] + graph[u_node][v];
            if (collective_weight < dist[v]) {
                dist[v] = collective_weight;
                parent[v] = u_node;
            }
        }
    }
}

```

4.3 Execution Profile & Routing Optimality

Because the graph contains a small number of vertices ($V = 7$), using a matrix loop configuration yields an execution profile of $O(V^2)$. This approach provides excellent predictability and low constant-factor performance overhead without the complexity of a binary min-heap structure.

Empirical Verification Matrix

Test Case Identifier	Input Pattern Sequence	Expected System Behavior	Actual Empirical Result	Status
NAV-01	Start Input: E	Triggers intercept: [Validation Error] 'E' is not approved.	Safe loop restart handled.	PASSED
NAV-02	Start Input: Engineering	Computes path to Dormitory: Distance = 8m. Path = Engineering -> Administration -> Dormitory.	Optimal path found.	PASSED

Test Case Identifier	Input Pattern Sequence	Expected System Behavior	Actual Empirical Result	Status
NAV-03	Start Input: exit	Breaks navigation loop and clears graph memory allocations.	Clean system exit.	PASSED

5. TELEMETRY DATA COMPRESSION UTILITY

5.1 Mathematical Foundations of Huffman Encoding

The Telemetry Data Compression Utility utilizes **Huffman Coding**, a classic greedy algorithm designed to generate prefix-free variable-length binary codes based on character frequency distributions. Characters that appear frequently in the text stream are assigned shorter bit configurations, while rare characters receive longer sequences. This optimizes the total storage footprint for text data.

5.2 Bitwise Serialization & Custom Header Architecture

To ensure the compressed payload can be decompressed standalone, the compressor writes a custom serialization header directly into the output file `telemetry.huff`. It includes a unique identifier, the unique character count, and the explicit character frequency dictionary entries, followed immediately by the bit-packed variable length codes padded out to the nearest byte boundary.

Core C Implementation: Low-Level Bit-Packing Stream Driver

```
void serialize_huffman_bit(FILE* target_file, int active_bit, unsigned char* bit_buffer,
int* tracking_bit_count) {
    // Inject active bit state into tracking byte frame shift orientation
    if (active_bit == 1) {
        *bit_buffer |= (1 << (7 - *tracking_bit_count));
    }
}
```

```

    (*tracking_bit_count)++;

    // Once exactly 8 bits are collected, pipe the raw byte to disk stream
    if (*tracking_bit_count == 8) {
        fputc(*bit_buffer, target_file);
        *bit_buffer = 0;           // Re-flush buffer content values
        *tracking_bit_count = 0;   // Reset tracking sequence counter
    }
}

```

5.3 Lossless Performance Statistics & Resource Teardown

For short input files, the header metadata can sometimes make the .huff file larger than the uncompressed text. This is expected behavior for Huffman encoding on small inputs. As file sizes increase, this fixed header overhead becomes negligible, resulting in significant file size savings. Data fidelity is validated by comparing the original file against telemetry_restored.txt byte-for-byte to confirm a perfect, lossless transformation. Finally, a post-order traversal function recursively deallocates the entire tree structure to prevent memory leaks.

Empirical Verification Matrix

Test Case Identifier	Input Pattern Sequence	Expected System Behavior	Actual Empirical Result	Status
HUFF-01	Run compression script	Generates telemetry.huff and prints compression metrics.	Header and binary file created.	PASSED
HUFF-02	Run decompression script	Rebuilds tree from header and outputs telemetry_restored.txt.	Accurate tree reconstruction.	PASSED

Test Case Identifier	Input Pattern Sequence	Expected System Behavior	Actual Empirical Result	Status
HUFF-03	Execute diff command	Output returns empty, verifying that both files are identical.	100% Lossless validation.	PASSED

6. OVERALL TESTING METHODOLOGY & VERIFICATION MATRICES

To guarantee high stability, the entire portfolio was compiled under the strict ISO C99 programming standard using the GNU Compiler Collection (gcc) with compiler warning flags enabled:

```
gcc -Wall -Wextra -std=c99 -O2
```

Memory correctness was validated using the valgrind memory analysis tool to ensure that all dynamically allocated structures across the five applications are freed correctly upon termination:

```
==18420== HEAP SUMMARY:
==18420==      in use at exit: 0 bytes in 0 blocks
==18420==    total heap allocs: 54 allocs, 54 frees, 12,482 bytes allocated
==18420==
==18420== All heap blocks were successfully freed -- no leaks are possible
```

7. CONCLUSION & ACADEMIC REFLECTION

This project highlights the importance of selecting and implementing the right data structures to build efficient, low-level software solutions. Restricting the capacity of a Doubly Linked List

creates a highly predictable memory footprint suitable for embedded logging pipelines. Implementing a text-based Binary Search Tree provides fast, efficient data validation while preserving memory sanitation. Using Adjacency Matrices simplifies graph mutations and speeds up network lookups to $O(1)$ time. For pathfinding, Dijkstra's algorithm offers predictable performance on small graphs, while Huffman coding demonstrates how custom serialization headers enable standalone, lossless file compression. Writing these foundational data structures natively in C reinforces a disciplined approach to managing memory allocation, understanding algorithmic efficiency, and building robust, error-tolerant software architectures.

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- ---

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